

COCOBOD

Ghana Cocoa Board — CHED

Telephone Exchange Management System

Complete System Documentation, Database Audit &
Production-Ready Enhancement Plan

Stack: PHP | React | MySQL 9.1 | phpMyAdmin 5.2

Database: cocobod_telephone_exchange | Generated: May 2026

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1. System Overview

1.1 What the System Does

The COCOBOD Telephone Exchange Management System (TEMS) is a live, operational web-based platform built specifically for the Cocoa Health and Extension Division (CHED) of the Ghana Cocoa Board. It digitises the full lifecycle of the COCOBOD internal telephone exchange — from logging every incoming and outgoing call, to managing telephone station faults, scheduling preventive maintenance, generating departmental billing invoices, and maintaining a complete audit trail of all system activity.

The system is currently running on a local development server (localhost) using PHP 8.3.14, MySQL 9.1.0, and React, with phpMyAdmin 5.2.1 as the database management interface. It has six active user accounts across three roles and contains real operational data including 23 call records, 8 fault reports, 8 billing invoices, 14 telephone stations, 8 departments, and 23 staff directory entries.

1.2 Technology Stack

Layer	Technology	Role in System
Frontend	React.js	Dynamic UI rendering, component state, real-time table updates
Backend	PHP 8.3.14	REST API endpoints, business logic, session auth, billing calculations
Database	MySQL 9.1.0	All data persistence — calls, faults, bills, users, stations, departments
DB Admin	phpMyAdmin 5.2.1	Database management interface for direct schema and data operations
Server	localhost / XAMPP	Local development environment — to be migrated to production server for COCOBOD deployment

1.3 User Roles & Active Accounts

The system currently has 6 active user accounts across 3 roles:

ID	Username	Email	Role	Name
1	admin	admin@cocobod.gov.gh	admin	System Administrator
2	receptionist	receptionist@cocobod.gov.gh	receptionist	Abena Mensah
3	technician	technician@cocobod.gov.gh	technician	Kwame Asante
4	jdoe	jdoe@cocobod.gov.gh	receptionist	John Doe
5	aboateng	aboateng@cocobod.gov.gh	technician	Ama Boateng
6	Lovelace	lovelacejohnkwakubaidoo@gmail.com	receptionist	Lovelace John Baidoo

⚠ A 4th role 'Manager' exists in the user_roles table but no users are currently assigned to it. This is an enhancement opportunity.

2. Actual Database Schema — Full Audit

The following is a complete audit of every table in the cocobod_telephone_exchange database, based on the actual SQL dump exported on 28 May 2026. This is the ground truth of the current system.

2.1 Complete Table Inventory

#	Table Name	Engine	Records	Purpose
1	users	InnoDB	6 active users	Authentication, roles, session management
2	departments	InnoDB	8 departments	COCOBOD department registry
3	stations	InnoDB	14 stations	Telephone desk stations with extensions and status
4	staff	MyISAM	23 staff	COCOBOD staff directory with phone extensions
5	employees	MyISAM	9 employees	System user employee profiles linked to users table
6	calls	MyISAM	23 call records	Full call log: caller, extension, department, type, duration, status
7	call_types	MyISAM	10 types	Internal / Local / National / International / Mobile with rates
8	call_transfers	MyISAM	0 records	Call transfer log: from/to department and employee
9	contacts	MyISAM	10 contacts	External contact directory (government, partners, individuals)
10	fault_reports	InnoDB	8 fault reports	Primary fault tracking: hardware/software/network, severity, status
11	fault_reporting	MyISAM	0 records	Legacy/secondary fault table with asset linkage — appears unused
12	fault_types	MyISAM	6 types	Fault classification: Line Dead, Poor Signal, Equipment, Network, etc.
13	maintenance_schedules	InnoDB	0 records	Scheduled preventive/corrective maintenance jobs (empty — not yet in use)
14	bills	MyISAM	8 invoices	Departmental telephone billing invoices with full financial breakdown
15	bill_items	MyISAM	0 records	Line items linking individual calls to bills — currently unpopulated

16	bill_types	MyISAM	8 types	Billing cycles: Daily / Weekly / Monthly / Annual
17	telecel_tariffs	MyISAM	8 tariffs	Telecel Standard and Premium call rate schedules in GHS per minute
18	ranks	MyISAM	16 ranks	Staff rank hierarchy (Director to Clerk) — duplicate sets exist
19	assets	MyISAM	0 records	Telephone equipment asset register — schema exists, no data yet
20	audit_logs	InnoDB	25 entries	System audit trail: login events and data changes with IP and user agent
21	notifications	MyISAM	0 records	In-app notification bell system — schema ready, not yet triggered
22	system_settings	InnoDB	10 settings	Configurable system parameters: org name, tax rate, session timeout, etc.
23	user_roles	MyISAM	4 roles defined	Role definitions table — Admin, Receptionist, Technician, Manager
24	system_logs	MyISAM	0 records	Secondary system event log — separate from audit_logs, currently unused

2.2 Departments (8 Active)

ID	Code	Department Name	Extension	Location	Description
1	ADMIN	Administration	100	Block A, Ground Floor	General Administration
2	HR	Human Resources	101	Block A, 1st Floor	HR & Personnel
3	FIN	Finance & Accounts	102	Block B, Ground Floor	Finance, Accounts & Procurement
4	IT	Information Technology	103	Block B, 1st Floor	IT & Telecommunications
5	HLTH	Health Services	104	Block C, Ground Floor	Occupational Health & Safety
6	EXT	Extension Services	105	Block C, 1st Floor	Cocoa Extension & Field Services
7	RND	Research & Development	106	Block D, Ground Floor	Cocoa Research & Development
8	QC	Quality Control	107	Block D, 1st Floor	Quality Assurance & Control

2.3 Telephone Stations (14 Stations)

ID	Station Name	Code	Dept	Extension	Status	Location
5	Reception Desk 1	STN-001	Admin	100	active	Block A, Reception
6	Reception Desk 2	STN-002	Admin	101	active	Block A, Reception
7	HR Manager Office	STN-003	HR	201	active	Block A, 1st Floor
8	Finance Manager	STN-004	Finance	301	active	Block B, Ground Floor
9	IT Department	STN-005	IT	103	active	Block B, 1st Floor
10	Health Services	STN-006	Health	401	 maintenance	Block C, Ground Floor
11	Extension Services	STN-007	EXT	501	active	Block C, 1st Floor
12	Conference Room A	STN-008	Admin	110	active	Block A, Conference Room
13	Director Office	STN-009	Admin	001	active	Block A, Director Suite
14	IT Technician Bench	STN-010	IT	104	 faulty	Block B, IT Workshop

 STN-006 (Health Services) is under maintenance. STN-010 (IT Technician Bench) is faulty. Both require technician attention.

2.4 Current Fault Reports (8 Reports)

Reference	Department	Type	Severity	Status	Description
FLT-2024-001	Health Services	Hardware	High	 Resolved	Desk phone not receiving calls — handset replaced
FLT-2024-002	IT (Station 10)	Hardware	Medium	 In Progress	IT bench phone completely dead — no dial tone
FLT-2024-003	Administration	Network	Critical	 Assigned	External line drops after 2-3 minutes from reception
FLT-2024-004	Administration	Hardware	Medium	 Open	Conference room speakerphone echo/feedback noise
FLT-2024-005	IT Dept	Software	High	 In Progress	VoIP softphone crashing after Windows update
FLT-2024-006	Extension Services	Hardware	Medium	 Open	Phone line has static noise affecting call quality
FLT-2024-007	Finance	Network	Low	 Resolved	Billing portal authentication error — credentials reset
FLT-2024-008	HR	Hardware	Low	 Open	HR Manager office phone screen cracked — display unreadable

2.5 Billing Summary (8 Invoices)

Invoice	Account	Period	Calls	Amount (GHS)	Status
INV-2024-001	Extension Services Dept	Jan 2024	45	1,265.63	☑ Paid — Bank Transfer
INV-2024-002	Finance & Accounts Dept	Jan 2024	32	900.00	☑ Paid — Cheque
INV-2024-003	Human Resources Dept	Jan 2024	28	787.50	☑ Paid — Bank Transfer
INV-2024-004	Health Services Dept	Jan 2024	20	562.50	⚠ OVERDUE — Unpaid
INV-2024-005	Extension Services Dept	Feb 2024	38	1,068.75	⌚ Pending
INV-2024-006	Finance & Accounts Dept	Feb 2024	25	703.13	⌚ Pending
INV-2024-007	Human Resources Dept	Feb 2024	22	618.75	📬 Sent
INV-2024-008	IT Department	Feb 2024	15	421.88	📄 Draft

2.6 System Settings (Configured Values)

Setting Key	Current Value	Description
org_name	COCOBOD CHED	Short organisation name displayed in UI
org_full_name	Cocoa Health and Extension Division	Full organisation name for reports and documents
default_currency	GHS	All billing amounts in Ghanaian Cedi
tax_rate	12.5%	VAT rate applied to all invoices
service_charge	5%	Service charge percentage on bills
session_timeout	480 minutes (8 hours)	Auto-logout after inactivity
billing_due_days	30 days	Days until a bill is due after generation
support_email	itsupport@cocobod.gov.gh	IT support contact email

3. Issues & Gaps Identified in Audit

The following issues were identified during the database audit. These are not critical failures — the system is operational — but they represent technical debt that should be addressed before a production deployment or COCOBOD presentation.

3.1 Critical Issues

Issue	Detail	Fix Required
Duplicate call_types data	call_types table has 10 records but only 5 are unique types — IDs 1-5 and 6-10 are duplicates with different codes and rates	Deduplicate — keep IDs 6-10 (have type codes). Update all calls.call_type_id references accordingly.
Duplicate ranks data	ranks table has 16 records but 8 are duplicates (IDs 1-8 and 9-16). Both sets are active.	Keep IDs 9-16 (have rank_level). Remove IDs 1-8 after verifying no employee FK references them.
Duplicate bill_types data	bill_types has 8 records but IDs 1-4 and 5-8 are duplicates of the same billing cycles	Keep IDs 5-8 (have type_code). Remove IDs 1-4 after verifying bills.bill_type_id references.
calls.call_date all null	All 23 call records have call_date = '0000-00-00' and call_time = '00:00:00'. Only start_time and end_time are properly populated.	Update call_date and call_time from start_time for all existing records. Enforce NOT NULL + default NOW() going forward.
bills.generated_date all null	All 8 bills have generated_date = '0000-00-00'. created_at is populated correctly.	Update generated_date from created_at for existing records. Enforce on insert going forward.
Mixed storage engines	System uses both InnoDB (users, departments, stations, fault_reports, maintenance_schedules, system_settings) and MyISAM (most other tables). This breaks cross-table transactions.	Convert all MyISAM tables to InnoDB for transactional integrity. Use: ALTER TABLE x ENGINE=InnoDB;

3.2 Enhancement Gaps

- bill_items table is empty — individual call charges are not being linked to invoices. Billing is calculated in aggregate, not per-call.
- call_transfers table is empty — transfer tracking is not yet operational despite the schema being ready.
- maintenance_schedules table is empty — preventive maintenance scheduling is built but not being used.
- assets table is empty — telephone equipment asset register exists in schema but has no data.
- fault_reporting table (separate from fault_reports) has 0 records — appears to be a legacy or duplicate table.
- notifications table has 0 records — in-app notification system exists but is not being triggered by any events.
- system_logs table has 0 records — secondary event log is not being written to.
- Manager role is defined in user_roles but no users are assigned to it.
- staff table (23 COCOBOD staff) has no department_id populated — all NULL. Cannot filter staff by department.

4. Dashboard Functionality — Current State

4.1 Administrator Dashboard

The Administrator has full system access. Based on the database schema and seed data, the following functions are confirmed operational:

4.1.1 User Management

- View all system users — currently 6 accounts across 3 roles
- Create new user accounts with role assignment (admin, receptionist, technician)
- Activate and deactivate user accounts (is_active flag)
- Password management — bcrypt hashing confirmed in users.password_hash
 - Note: User 6 (Lovelace) uses \$2b\$ prefix (bcrypt) while users 1-5 use \$2a\$ — both are valid bcrypt variants

4.1.2 Department & Station Management

- View and manage 8 COCOBOD departments with codes, locations, and extensions
- View and manage 14 telephone stations — including status tracking (active/maintenance/faulty)
- Station STN-006 (Health Services) shows status: maintenance
- Station STN-010 (IT Technician Bench) shows status: faulty

4.1.3 Billing Management

- Generate invoices for departments based on call logs
- 8 invoices generated across 4 departments — Jan and Feb 2024
- Tax rate: 12.5% (VAT) applied to all invoices — configured in system_settings
- Bill statuses: Paid, Overdue, Pending, Sent, Draft, Cancelled
- Currency: GHS (Ghanaian Cedi) — confirmed in system_settings
- Invoice INV-2024-004 (Health Services) is OVERDUE — GHS 562.50 unpaid

4.1.4 System Configuration

- 10 configurable system settings managed via system_settings table
- Settings include: org name, tax rate, service charge, session timeout, billing due days, support contacts
- All settings are editable via the admin panel

4.1.5 Audit Trail

- audit_logs table is active with 25 entries covering: LOGIN events, CREATE_USER, CREATE_CALL, GENERATE_BILL, UPDATE_FAULT, UPDATE_CALL
- Logs capture: user_id, action, table_name, record_id, ip_address, created_at, old_values (JSON), new_values (JSON), user_agent
- Most recent activity: User 6 (Lovelace) logged in on 2026-05-27

4.2 Receptionist Dashboard

The Receptionist handles the front-line telephone exchange operations. Two receptionists are active: Abena Mensah and John Doe. A third (Lovelace Baidoo) was added on 27 May 2026.

4.2.1 Call Logging

- 23 call records logged — referenced as CHED-2024-001 through CHED-2024-023
- Call types: Internal, Local, National, International — rates from call_types table
- Priority levels: low / medium / high / urgent — all four in use
- Call statuses: Completed (18), Missed (1), Transferred (1), Busy (1), Pending remainder
- Follow-up tracking: calls with follow_up_required=1 (7 calls) have follow_up_date set
- Calls are linked to: contacts, departments, employees, stations

4.2.2 Contact Directory

- 10 external contacts in the contacts table: government bodies, partners, and individuals
- Types: government (5), partner (2), individual (2), external (1)
- Key contacts: COCOBOD HQ, Kumasi Regional Office, Takoradi Port Authority, Produce Buying Company, Cocoa Processing Co., Standard Chartered Bank

4.2.3 Fault Report Submission

- Receptionist (user_id 2 and 8) submitted most of the 8 fault reports
- Reports include: fault type, category, description, severity, affected station and department
- Fault categories: equipment, connection, phone_line, billing

4.3 Technician Dashboard

Two technicians are active: Kwame Asante (senior) and Ama Boateng. Both are in the IT department (department_id 4).

4.3.1 Fault Queue Management

- View all 8 fault reports — filterable by status: open (4), in_progress (2), assigned (1), resolved (2)
- Fault priority breakdown: Critical (1), High (2), Medium (4), Low (1)
- 3 critical/high faults currently unresolved: FLT-2024-002 (IT bench), FLT-2024-003 (Reception line drops), FLT-2024-005 (VoIP crashing)

4.3.2 Status Updates

- Technician can update fault_status: open → assigned → in_progress → resolved → closed
- Resolution notes field records what was done — confirmed populated on resolved faults
- estimated_cost and actual_cost tracked per fault (estimated only — actuals not yet populated)

4.3.3 Maintenance Scheduling

- maintenance_schedules table is ready with full schema: preventive / corrective / upgrade types
- Currently 0 scheduled jobs — this feature exists in the schema but is not yet being used

💡 *Activating the maintenance scheduling feature would be a strong demonstration point for COCOBOD.*

5. Production-Ready Enhancement Plan

The system is functional and contains real operational data. The following enhancements will transform it from a working prototype into a production-ready enterprise platform suitable for a COCOBOD board-level presentation.

5.1 Priority 1 — Data Integrity Fixes (Do First)

 These fixes are essential before any presentation. They correct the database issues identified in Section 3.

Fix	SQL Action Required	Impact
Deduplicate call_types	Delete IDs 1-5, update all FK references in calls table to use IDs 6-10	Fixes rate calculations
Deduplicate ranks	Delete IDs 1-8, verify employees.rank_id uses IDs 9-16	Cleans staff profiles
Fix calls.call_date	UPDATE calls SET call_date=DATE(start_time), call_time=TIME(start_time) WHERE call_date='0000-00-00'	Enables date filtering
Convert to InnoDB	ALTER TABLE calls, contacts, staff, employees, fault_reports... ENGINE=InnoDB	Enables transactions
Populate staff.dept	Update staff table department_id based on job title mapping (23 staff, all currently NULL)	Enables dept filtering
Link bill_items	Populate bill_items by linking calls to bills based on department_id and billing period	Enables per-call billing

5.2 Priority 2 — Activate Ready-Built Features

The following features are already built in the database schema but are not yet active. Activating them requires minimal development effort and creates major demo impact.

- Maintenance Scheduling — maintenance_schedules table is ready. Build the UI for the admin to schedule preventive maintenance and assign to technicians. This directly addresses the faulty stations (STN-006 and STN-010).
- Call Transfers — call_transfers table is ready. Build the receptionist 'Transfer Call' action that logs from/to department and employee. Show transfer history per call.
- Notification Bell — notifications table is ready. Wire it to: new fault reports, fault status changes, new bills generated, overdue bill alerts. Show the unread count badge on the top bar.
- Asset Register — assets table is ready. Populate with COCOBOD's telephone equipment. Link assets to stations and fault reports. This creates a full equipment lifecycle view.
- Manager Role — the user_roles table has a Manager role. Create one manager account for the demo and build a read-only reporting dashboard for them.

5.3 Priority 3 — UI/UX Enhancements

5.3.1 Dashboard KPI Cards (Admin)

The admin dashboard should show real-time KPI cards pulled directly from the actual data:

- Total Active Stations: 12 active / 1 maintenance / 1 faulty (from stations table)
- Open Faults: 4 open, 1 critical (from fault_reports table)
- Calls Today / This Week / This Month (from calls table, grouped by start_time date)
- Outstanding Bills: GHS 562.50 overdue (INV-2024-004 Health Services)
- Total Calls by Type: breakdown of Internal / Local / National / International

5.3.2 Advanced Table Features

- All data tables must support: column sorting, inline search, pagination (10/25/50 rows), CSV export
- Status columns must show colour-coded badges matching the fault/bill/call status values in the database
- Fault table: Critical = red badge, High = orange, Medium = yellow, Low = grey
- Bill table: Overdue = red, Pending = amber, Paid = green, Draft = grey, Sent = blue
- Station table: active = green dot, maintenance = amber dot, faulty = red dot

5.3.3 Telecel Tariff Integration

The telecel_tariffs table has 8 active tariff rates (Standard and Premium tiers). These should be:

- Displayed in the admin settings panel as an editable tariff table
- Used as the calculation basis for call billing (currently call_types has rates — reconcile with telecel_tariffs)
- Standard Local: GHS 0.15/min | National: GHS 0.25/min | International: GHS 2.50/min | Mobile: GHS 0.20/min

5.4 Priority 4 — Security Hardening

- Session timeout is 480 minutes (8 hours) — reduce to 60 minutes for security
- Enforce CSRF tokens on all POST forms — especially billing and user management
- Add rate limiting on login: lock after 5 failed attempts (per audit_logs.LOGIN pattern)
- Validate that all PHP endpoints check session role before processing — prevent URL-based access bypass
- Ensure audit_logs captures old_values and new_values JSON on all data changes — currently NULL for most entries

6. Presentation Guide for COCOBOD

6.1 Key Statistics to Quote in Your Presentation

Use these real figures from the actual database in your slides and demo:

Metric	Actual Value (from database)
Total call records logged	23 calls (CHED-2024-001 to CHED-2024-023)
Departments covered	8 departments — Admin, HR, Finance, IT, Health, Extension Services, R&D, QC
Telephone stations managed	14 stations across all COCOBOD blocks (A, B, C, D)
Active fault reports	8 reports — 1 critical, 2 in progress, 4 open, 2 resolved
Total invoices generated	8 invoices — GHS 6,328.13 total billed
Outstanding/Overdue billing	GHS 562.50 — Health Services Dept (INV-2024-004, Jan 2024)
Audit trail entries	25 logged events — logins, user creation, calls, billing, fault updates
COCOBOD staff in directory	23 named staff from Executive Director to Principal Security Guard
External contacts tracked	10 contacts — COCOBOD HQ, Port Authority, Produce Buying Co., Standard Chartered, etc.

6.2 Recommended 12-Slide Presentation Structure

#	Slide Title	Content	Duration
1	Title Slide	COCOBOD TEMS — CHED PHP + React + MySQL Your name May 2026	30 sec
2	The Problem	Manual telephone exchange: lost call records, no fault tracking, no billing accountability, no audit trail	1 min
3	The Solution	TEMS: digital call logging, fault management, billing, staff directory, and full audit trail — web-based, role-based access	1 min
4	System Architecture	Browser → React frontend → PHP REST API → MySQL 9.1 3 roles, 24 tables, real COCOBOD data	1 min
5	Live Data Snapshot	Quote real stats: 23 calls, 8 faults, GHS 6,328 billed, 14 stations, 8 departments — all live in the system	1 min
6	Admin Dashboard Demo	Login as admin — show KPI cards, user management, billing overview, system settings	3 min
7	Receptionist Demo	Log in as receptionist — log a call, look up an extension, submit a fault report for faulty STN-010	2 min
8	Technician Demo	Log in as technician — view fault queue, update FLT-2024-002 to resolved, add resolution notes	2 min

9	Billing Demo	Show INV-2024-004 (overdue), explain tax calculation (12.5% VAT), show paid vs pending status	1.5 min
10	Security & Audit	Show audit_logs: 25 entries, IP addresses, action types — full accountability trail	1 min
11	Enhancement Roadmap	Asset register, maintenance scheduling, call transfer tracking, notification system, manager role	1 min
12	Q&A	Prepared answers: backup strategy, offline capability, PABX integration, scalability to other COCOBOD divisions	5 min

6.3 Likely COCOBOD Questions — Prepared Answers

- 'Can it work for other COCOBOD divisions beyond CHED?' — Yes. The department structure is fully configurable. Any division can be added in system_settings with their own departments and staff.
- 'How is data backed up?' — MySQL supports scheduled dumps via cron jobs. For production deployment, we recommend daily automated backups to a secure offsite server.
- 'Can it integrate with the existing PABX system?' — The calls table has all fields needed for PABX CDR (Call Detail Record) import. The API can be extended to receive records directly from a PABX via webhook.
- 'What happens if internet goes down?' — The system runs on your local server infrastructure, not the public internet. As long as the COCOBOD internal network is up, the system works.
- 'How many users can it support?' — The current schema supports unlimited users. Performance testing recommended before deployment to more than 50 concurrent users.

6.4 Demo Data Tips

- The system already has 23 realistic CHED call records with actual COCOBOD contacts (Produce Buying Company, Takoradi Port Authority, Kumasi Regional Office, Standard Chartered Bank).
- Fault FLT-2024-003 (Critical — reception line dropping calls) is a perfect live demo: realistic, urgent, unresolved. Show it being assigned and updated.
- Invoice INV-2024-004 (Health Services overdue) is a strong billing demo: show how the system tracks accountability.
- Staff directory has 23 real COCOBOD staff names and extension numbers — from Executive Director (ext. 490) to Principal Security Guard (ext. 497).

COCOBOD Telephone Exchange Management System — v2.0

Based on actual database export — cocobod_telephone_exchange | May 28, 2026 | CONFIDENTIAL